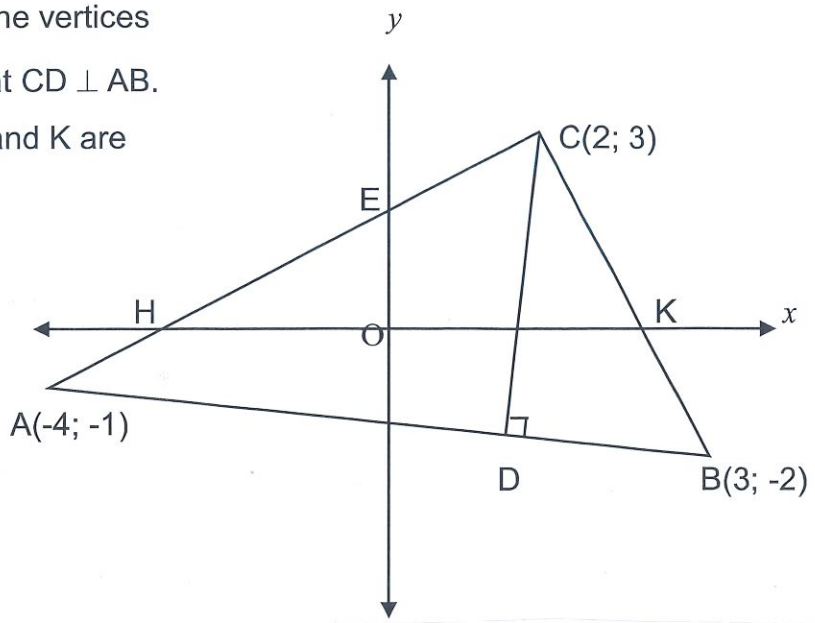
GRADE 10 Memo **Consolidation worksheet**

1. A(-4; -1), B(3; 2) and C(2; 3) are the vertices of ΔABC. D is a point of AB so that $CD \perp AB$. E is a point on the y-axis and H and K are on the x-axis. Determine:

   a) the equation of AC
   b) the equation of BC
   c) the equation of CD
   d) the length of OE
   e) the length of HK



1 a) $m = \frac{3-(-1)}{2-(-4)} = \frac{2}{3}$

$y = \frac{2}{3}x + c$ sub C (2;3)

$3 = \frac{2}{3}(2) + c$

$\frac{5}{3} = c$

Eqn AC: $y = \frac{2}{3}x + \frac{5}{3}$

b) $m_{BC} = \frac{3-(-2)}{2-3}$

$m_{BC} = \frac{5}{-1}$

$y = -5x + c$ sub C

$3 = -5(2) + c$

$13 = c$

Eqn: BC $y = -5x + 13$

c) $CD \perp AB$

$\therefore m_1 \times m_2 = -1$

$m_{AB} = \frac{-2-(-1)}{3-(-4)}$

$m_{AB} = -\frac{1}{7}$

$\therefore m_{CD} = 7$ ($\perp$)

$\therefore y = 7x + c$ sub C (2;3)

$3 = 7(2) + c$

$-11 = c$

$\therefore y = 7x - 11$

d) E is y-int of line AC

$y = \frac{2}{3}x + \frac{5}{3}$

$\therefore OE = \frac{5}{3}$ units

e) H and K are x-intercepts

H is on AC

$0 = \frac{2}{3} + \frac{5}{3}$

$-\frac{5}{3} = \frac{2}{3}x$

$-\frac{5}{2} = x$ H $(-\frac{5}{2}; 0)$

P.T.O

e) continued.

K is on line BC

$0 = -5x + 13$

$5x = 13$

$x = 13/5$ K $(\frac{13}{5} ; 0)$

Length HK

$= 5/2 + 13/5$

$= \frac{51}{10}$

$= 5,1$ units

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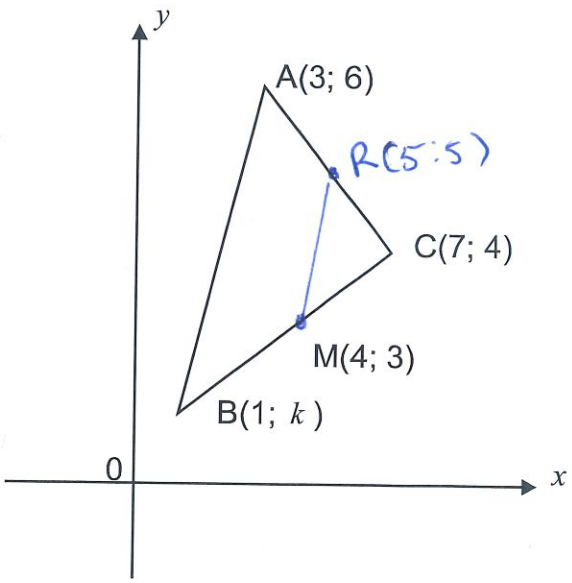
2. In the figure, A(3; 6) , B(1; k) and C(7; 4) are the vertices of a triangle with M(4; 3) the midpoint of BC.

a) Show that $k = 2$.

b) If R is the midpoint of AC, show using analytical methods that:

i) MR = ½ BA

ii) MR || BA



2. a) Midpt M

$(4 ; 3) = (\frac{1+7}{2} ; \frac{k+4}{2})$

$\therefore 3 = \frac{k+4}{2}$

$6 = k + 4$

$2 = k$

b) i) $MR = \sqrt{(4-5)^2 + (3-5)^2}$

$= \sqrt{5}$

R $= (\frac{3+7}{2} ; \frac{6+4}{2})$

R $= (5 ; 5)$

$AB = \sqrt{(3-1)^2 + (6-2)^2}$

$= 2\sqrt{5}$

$\therefore AB = 2 \times MR$

ie $MR = 1/2\, AB$

2 b) ii $m_{AB} = \frac{6-2}{3-1} = \frac{4}{2} = 2$

$m_{MR} = \frac{5-3}{5-4} = \frac{2}{1}$

$\therefore m_{AB} = m_{MR}$

$\therefore$ AB // MR

3. In the figure, A(2; 1), B(8; 5) and C(4; 8) are the vertices of a triangle. The point M(x; y) lies on the y-axis such that CM is parallel to BA.

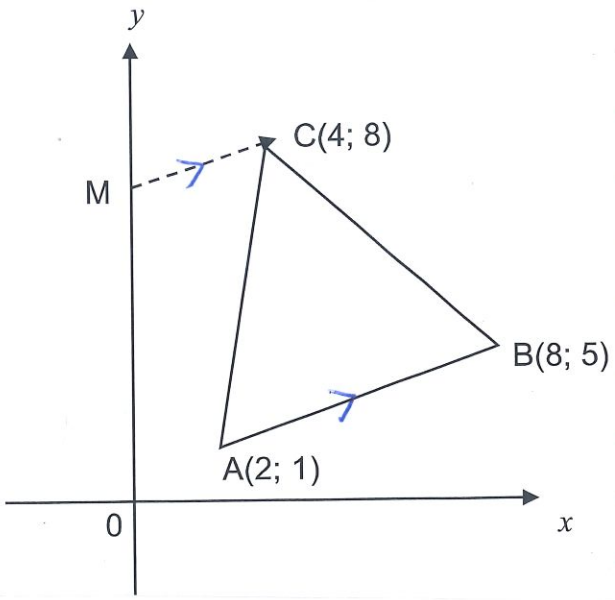
Determine the following:

a) the length of BC

b) the gradient of AB

c) the equation of CM, in the form y=mx + c

d) the co-ordinates of M



a) $BC = \sqrt{(8-4)^2 + (5-8)^2}$

$= \sqrt{4^2 + (-3)^2}$

$= \sqrt{25}$

BC = 5 units

b) $m_{AB} = \frac{5-1}{8-2} = \frac{2}{3}$

c) $m_{CM} = m_{AB}$

(Given MC // AB)

$m_{CM} = 2/3$

$y = 2/3\,x + c$ (sub C)

$8 = 2/3\,(4) + c$

$\frac{16}{3} = c$

$\therefore$ Eqn CM: $y = \frac{2}{3}x + \frac{16}{3}$

d) M is y-intercept of line CM

$\therefore$ M $(0; \frac{16}{3})$

4. The equation of a given straight line is $y = px + 6$.

Determine the value of p in each of the following cases:

a) the given line [$y = px + 6$] passes through the point (-1; -4).

b) the given line is // to the line $2y + x = 4$.

c) the given line is // to the x-axis.

a) $y = px + 6$ sub (-1 ; -4)

$-4 = p(-1) + 6$

$-10 = -p$

$10 = p$

b) $2y + x = 4$

$2y = -1x + 4$

$y = -\frac{1}{2}x + 2$

$m_1 = -1/2 \therefore m_2 = -1/2$ (lines are //)

$p = -1/2$

c) $y = px + 6$ is // to x-axis so it is a horizontal line with zero gradient

$\therefore p = 0$

5. Find the equation of the line passing through (-1; 1) and perpendicular to the line $x - 3y = 5$.

$x - 3y = 5$

$-3y = -1x + 5$

$y = 1/3\,x - 5/3$

$\therefore m_1 = 1/3$

$m_2 = -3$ because lines are ⊥ ($m_1 \times m_2 = -1$)

$y = -3x + c$ sub (-1 ; 1)

$1 = -3(-1) + c$

$1 = 3 + c$

$-2 = c \quad \therefore y = -3x - 2$

6. M(8 ½ ; -1) is the midpoint of line segment xy with $x(4a; b)$ and $y(5; -3)$. Determine the values of a and b.

x (4a; b) — M (8½; -1) — y (5; -3)

$$M = \left(\frac{x_1 + x_2}{2} ; \frac{y_1 + y_2}{2}\right)$$

$$\left(8\tfrac{1}{2} ; -1\right) = \left(\frac{4a + 5}{2} ; \frac{b - 3}{2}\right)$$

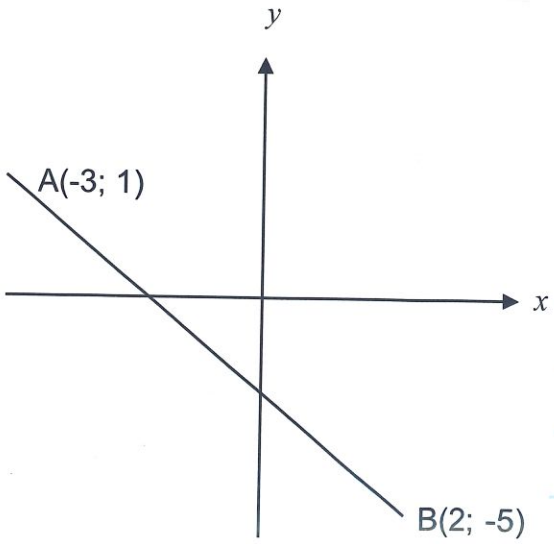
$$8\tfrac{1}{2} = \frac{4a+5}{2} \qquad -1 = \frac{b-3}{2}$$

$$17 = 4a + 5 \qquad -2 = b - 3$$

$$12 = 4a \qquad 1 = b$$

$$3 = a$$

7. In the sketch the points A(-3; 1) and B(2; -5) are indicated in a Cartesian plane.
The points C(m; n) and D(1; p) are given, but are not indicated.



a) If C is the midpoint of AB, calculate the value of m and n.

b) If A, B and D are collinear, calculate the value of p.

c) Determine the equation of the straight line which is parallel to AB and which passes through the origin.

a) $C(m; n) = \left(\frac{-3 + 2}{2} ; \frac{1 - 5}{2}\right)$

$$C(m; n) = \left(-\tfrac{1}{2} ; -2\right)$$

$m = -1/2$ and $n = -2$

7b) $m_{AB} = \frac{1-(-5)}{-3-2} = \frac{6}{-5}$

Given: D (1 : p)

$m_{AD} = \frac{1-p}{-3-1} = \frac{1-p}{-4}$

But A, B, D collinear $\therefore m_{AB} = m_{AD}$

$\frac{1-p}{-4} = \frac{6}{-5}$

$-5(1-p) = -24$

$-5 + 5p = -24$

$5p = -19$

$p = \frac{-19}{5}$

c) $y = mx + c$ // AB $(m_1 = m_2)$

$\therefore y = -\frac{6}{5}x + c$ sub (0 ; 0)

$0 = -\frac{6}{5}(0) + c$

$0 = c$ $\therefore$ Eqn: $y = -\frac{6}{5}x$

8. In the figure, the co-ordinates of P are (2; -1).
The equation of QR is $x - 2y + 6 = 0$ and the equation of PR is given by $3x - y - 7 = 0$.

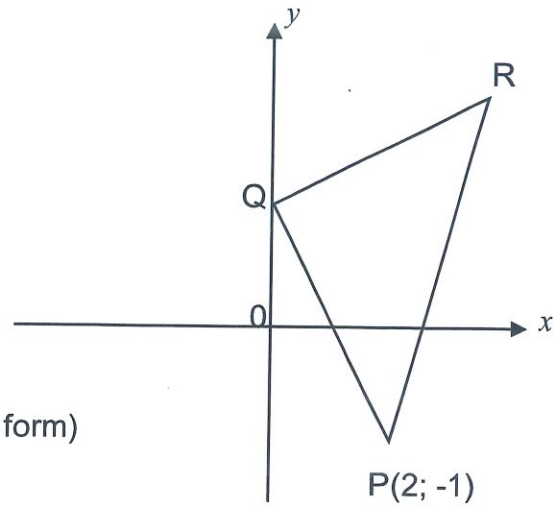
a) What is the gradient of QR?

b) Find the co-ordinates of Q.

c) Prove that $P\hat{Q}R = 90°$.

d) Determine the co-ordinates of R.

e) Find the length of PR. (leave answer in surd form)



8 QR: $x - 2y + 6 = 0$
$-2y = -1x - 6$
$y = \frac{1}{2}x + 3$ } Given

PR: $3x - y - 7 = 0$
$-y = -3x + 7$
$y = 3x - 7$ } Given

a) $m_{QR} = 1/2$

b) Q is y-intercept of QR
$\therefore$ Q (0;3)

c) $m_{QP} = \frac{3 - (-1)}{0 - 2} = \frac{4}{-2} = -2$

$m_{QR} = \frac{1}{2}$ See above

$\therefore m_{QP} \times m_{QR} = -2 \times 1/2$
$= -1$

$\therefore QP \perp QR$
ie $P\hat{Q}R = 90°$

d) To find R, solve simultaneously lines QR and PR
$y = 1/2x + 3$ & $y = 3x - 7$
$1/2x + 3 = 3x - 7$ (Put $y_1 = y_2$)
$10 = 2\frac{1}{2}x$
$4 = x$

$\therefore$ Sub into PR: $y = 3(4) - 7$
$y = 5$

$\therefore$ R (4;5)
Pt of intersection of the lines

8 (e) $PR = \sqrt{(2-4)^2 + (-1-5)^2}$

$= \sqrt{(-2)^2 + (-6)^2}$

$= \sqrt{40}$

$\underline{PR = 2\sqrt{10}}$